YouTube Music on Raspberry Pi: Setup Guide

A phone-controlled YouTube Music player for the Pi (ytm.py): search, playlists, queue, downloads, local folders and local playlists, played through mpv. This version lists every dependency and setting.

What you get

- A phone page at `http://<pi-ip>:8001`: search songs, videos, albums and playlists, or open any public playlist by link.
- A queue you can reorder and trim, favourites, saved playlists and Auto Play after a reboot.
- Downloads as Opus into the folder you last selected, plus a Local tab to browse folders and build local playlists.
- System volume buttons (the same method as radio.py) with a peak limiter against distortion.
- No Google login or cookies needed.

Requirements

- A Raspberry Pi (built and tested on a Pi 5) running Raspberry Pi OS Bookworm or newer, desktop edition, which uses PipeWire for audio.
- Python 3.9 or newer (Bookworm ships 3.11).
- Internet access, working audio output, and a phone on the same network as the Pi.
- The `ytm.py` file.

Dependencies

Component	How to install	What it is for
mpv	apt	Plays streams and local files; the app controls it over a socket
yt-dlp	pip	Resolves YouTube streams for mpv and downloads songs
ytmusicapi	pip	Search and public playlist lookup, no login

Component	How to install	What it is for
Flask	pip	Serves the phone page and the API
ffmpeg	apt	Writes tagged `.opus` files for downloads (without it, downloads are `.webm`)
nodejs	apt	JavaScript runtime that yt-dlp may need for YouTube
python3-pip	apt	Installs the pip packages
PipeWire with wpctl (wireplumber)	Preinstalled on the desktop edition	System volume and mute buttons
pkill (procps)	Preinstalled	Clears a stale mpv when the app starts

Raspberry Pi OS Lite has no PipeWire. The volume buttons then fall back to mpv's own volume, which still works.

1. Install everything

```
sudo apt update && sudo apt full-upgrade -y
sudo apt install -y mpv ffmpeg python3-pip nodejs
pip install -U flask ytmusicapi yt-dlp --break-system-packages
echo 'export PATH=$HOME/.local/bin:$PATH' >> ~/.bashrc && source ~/.bashrc
```

Check the tools:

```
mpv --version | head -1
yt-dlp --version
which -a yt-dlp
python3 -c "import flask, ytmusicapi; print('python modules ok')"
wpctl status
```

- `which -a yt-dlp` should list only `~/.local/bin/yt-dlp`. If `/usr/bin/yt-dlp` also appears, remove the old copy with `sudo apt remove yt-dlp`.
- If yt-dlp warns that it needs a JavaScript runtime or a newer one, install Deno (or a newer Node) and update yt-dlp again.
- If `wpctl` is not found, the app still works with mpv's own volume.

2. Copy the app

```
mkdir -p ~/YTMusic/DownloadedSongs
```

From your PC (change `pi` and the address to match your Pi):

```
scp ytm.py pi@<pi-ip>:~/YTMusic/
```

Find the Pi's address with `hostname -I`. The app creates its folders and data files by itself on first run.

3. Configure and test audio

```
wpctl status
wpctl set-default <sink-id>
wpctl set-volume -l 1.0 @DEFAULT_AUDIO_SINK@ 75%
mpv --no-video "https://www.youtube.com/watch?v=dQw4w9WgXcQ"
```

- Pick the output (headphones, DAC or HDMI) with `wpctl set-default`, using the id listed under Sinks in `wpctl status`.
- Hearing the track proves that audio, mpv and yt-dlp all work. Press `q` to quit.
- Keep the system volume around 70 to 80% for headroom. The app caps volume at 100% and adds a peak limiter.
- The app sets the audio session variables itself, so it also works when started from cron or SSH.

4. First run

```
cd ~/YTMusic && python3 ytm.py
```

On your phone, open `http://<pi-ip>:8001`. Press `Ctrl+C` on the Pi to stop it. Raspberry Pi OS has no firewall by default; if you turned one on, allow TCP port 8001 from your home network only.

5. Start automatically on boot

```
(crontab -l 2>/dev/null | grep -v ytm.py; echo "@reboot sleep 40 && python3
/home/pi/YTMusic/ytm.py > /home/pi/YTMusic/ytm.log 2>&1") | crontab -
```

- Change `/home/pi` if your username is different.

- The 40-second delay gives the audio service time to start.
- The `grep -v` removes any older ytm.py line, so running this twice does not create duplicates.
- Check with `crontab -l`, then reboot once to confirm the page comes up and plays.
- To start it now without rebooting: `nohup python3 ~/YTMusic/ytm.py > ~/YTMusic/ytm.log 2>&1 &`

Configuration reference

Set variables in front of `python3` in the command or cron line, for example `YTM_NORMALIZE=1 python3 ...`.

Setting	Where	Purpose
`MPV_AUDIO_DEVICE`	Environment	Force an audio device, for example `pipewire`. List names with `mpv --audio-device=help`.
`YTM_NORMALIZE=1`	Environment	Even out loudness between tracks (off by default).
`YTM_PORT`	Environment	Use a different port than 8001. radio.py uses 8000.
`YTM_DIR`	Environment	Use a different data folder than `~/YTMusic`. `DownloadedSongs` lives inside it.
`cookies.txt`	`~/YTMusic/cookies.txt`	Optional. If the file exists the app passes it to yt-dlp. Leave it out unless YouTube keeps blocking you, and use only a throwaway account.

Using the app

- **Search:** songs, videos, albums or playlists. Tap a result to play it and the rest of the list follows. The plus button adds to the queue, the heart saves a favourite, the down arrow downloads it.
- **Playlists:** paste a public YouTube playlist or song link, then Play all, Shuffle, Save, or download all.

- **Local, Songs view:** browse `DownloadedSongs` with a breadcrumb. Tap a folder to open it; its play button plays the whole folder. All lists every song under the current folder, Folders returns to the folder view, and New folder creates and selects a subfolder. Filter, Play, Shuffle and Add all to playlist work on what is on screen. The three-dot menu on a song offers Add to playlist, Add to queue and Delete from the Pi.
- **Local, Playlists view:** create, play, rename and delete playlists of local songs. Inside a playlist, the three-dot menu removes a song from it without deleting the file.
- **Favs:** your favourite tracks.
- **Queue:** tap a song to jump to it, use the up and down arrows to reorder and the cross to remove it. The Auto Play switch resumes the last queue after a reboot.
- **Player bar:** previous, play/pause, next, seek, and volume minus and plus (5% steps), slider and mute. Volume controls the Pi's system volume.

Downloads and local music

- Downloads are saved in the folder you last selected in the Local tab. The tab shows it as "New downloads save to". Each download keeps the folder it was queued with, even if you browse elsewhere afterwards.
- The app takes YouTube's best public audio and prefers native Opus, with no re-encoding. Files are named `Artist - Title [id].opus`. Without ffmpeg they are saved as `.webm` (Opus inside), which still plays.
- Downloads run one at a time. A song that already exists in any folder is skipped.
- To add your own music, copy files (opus, webm, m4a, mp3, flac, ogg, wav, aac, mka, wma) into `DownloadedSongs` or its subfolders, then open the Local tab. Folders that start with a dot are hidden.

Files and folders

Path	What it is
`~/YTMusic/ytm.py`	The app
`~/YTMusic/DownloadedSongs/`	Downloaded and local songs, with any subfolders
`~/YTMusic/favourites.json`	Favourite tracks
`~/YTMusic/saved_playlists.json`	Saved YouTube playlists
`~/YTMusic/local_playlists.json`	Playlists of local songs
`~/YTMusic/state.json`	Last queue, Auto Play setting and download folder

Path	What it is
`~/YTMusic/mpv.log`	mpv messages, rewritten at each start; useful when tracks skip
`~/YTMusic/ytm.log`	App messages
`~/YTMusic/cookies.txt`	Optional cookies, normally absent

Maintenance

```
# update the tools (about once a week)
pip install -U yt-dlp ytmusicapi --break-system-packages

# restart the app (also after copying a new ytm.py), then reload the page
pkill -f ytm.py
nohup python3 ~/YTMusic/ytm.py > ~/YTMusic/ytm.log 2>&1 &

# check that it is running and read recent mpv messages
pgrep -af ytm.py
tail -40 ~/YTMusic/mpv.log

# back up playlists, favourites and songs
tar czf ~/ytmusic-backup.tgz -C ~ YTMusic
```

Keeping yt-dlp and ytmusicapi current fixes most breakage, since YouTube changes often.

Troubleshooting

Problem	Likely cause	Fix
Tracks skip instantly	No audio output; `mpv.log` shows `Host is down`	Check `wpctl status`, set `MPV_AUDIO_DEVICE`, raise the cron delay to 60 seconds. With no desktop login at boot, run `sudo loginctl enable-linger pi`.
"Sign in to confirm you're not a bot" or HTTP 429	YouTube is throttling your IP	Wait a few hours, update yt-dlp, try another connection such as a phone hotspot, and stay off VPNs.
"The page needs to be reloaded"	An expired `cookies.txt`	Rename or delete `~/YTMusic/cookies.txt` and restart the app.

Problem	Likely cause	Fix
Search shows an error	ytmusicapi is outdated	Update it with pip (see Maintenance).
Warning about a JavaScript runtime	No runtime installed	`sudo apt install nodejs`, or install Deno.
Volume buttons do nothing	`wpctl` missing or no default output	Run `wpctl status`. Without wpctl the app uses mpv's own volume.
A download fails	See the message on the Local tab	Update yt-dlp and check free space with `df -h`.
Downloads are `.webm`	ffmpeg is not installed	`sudo apt install ffmpeg`. Only new downloads change.
A download went to an unexpected folder	It saves to the last folder selected in Local	Open the folder you want in the Local tab first; the tab shows the target.
A new button is missing or the page looks old	The old file is still running	Restart the app after copying the new `ytm.py`, then reload the page.
Page does not load	App not running, wrong network or port in use	Run `pgrep -af ytm.py`, read `ytm.log`, use the same network as the Pi, or set `YTM_PORT`.
A song is missing from a local playlist	The file was moved or deleted	Missing files are skipped; add the song again from its new location.

Good practice

- Run without cookies. They are only worth adding if you get persistent blocks, and then only from a throwaway Google account, never your main one. Exported cookies also expire quickly.
- Keep the load light: one stream at a time, no bulk downloading, and tools kept up to date.
- yt-dlp and ytmusicapi are unofficial tools. Use this for personal listening at home.
- The page has no password. Use it on your home network only and do not expose port 8001 to the internet.